Contents

Object Oriented Programming Languages

Complete Topic List

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- Instance and static members
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- Program structure, packages, and compilation units

Chapter Files

- `chapters/chap1_classes_objects.md`
- `chapters/chap2_encapsulation_members.md`

- chapters/chap3_polymorphism_types.md
- chapters/chap4_memory_management.md
- chapters/chap5_program_structure.md # Chapter 1: Classes and Objects

1.1 Classes and Objects, Instantiation

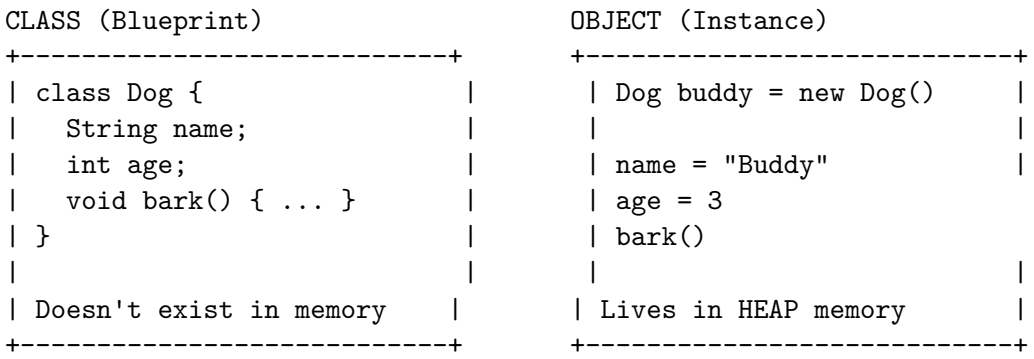
Definition / Theoretical Background

Class: A blueprint or template that defines the structure and behavior of objects. It specifies: - **Fields** (attributes/properties): Data that objects store - **Methods** (behaviors): Operations that objects can perform

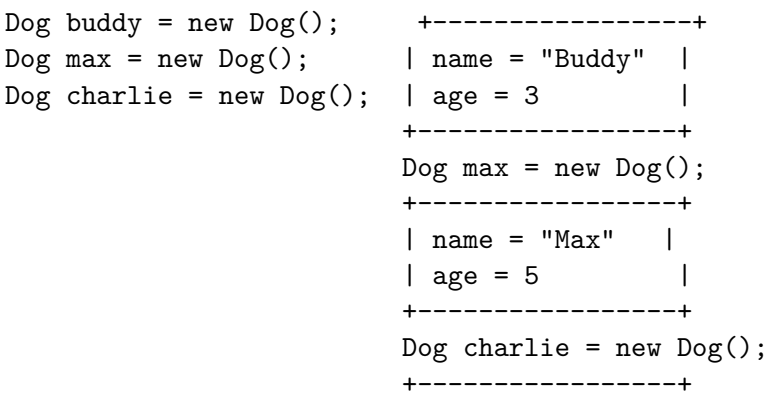
Object: An instance of a class - a concrete entity created from the blueprint.

Concept	What It Is	Example
Class	Blueprint	`class Dog { ... }`
Object	Instance of class	`Dog buddy = new Dog();`
Instantiation	Creating object from class	`new Dog()`
State	Values in object's fields	`buddy.name = "Buddy"`
Identity	Unique identifier	Each object has unique address
Behavior	Methods object can perform	`buddy.bark()`

Class vs Object Visualization



You can create MANY objects from ONE class!



```
                    | name = "Charlie"|
                    | age = 2         |
                    +-----------------+
```

Declaration and Instantiation

```
// Class definition
class Dog {
    // Instance variables (fields)
    String name;
    int age;

    // Constructor
    Dog(String name, int age) {
        this.name = name;
        this.age = age;
    }

    // Methods
    void bark() {
        System.out.println(name + " says Woof!");
    }
}

// Instantiation and use
public class Main {
    public static void main(String[] args) {
        Dog buddy = new Dog("Buddy", 3);  // Create object
        Dog max = new Dog("Max", 5);

        buddy.bark();  // "Buddy says Woof!"
        max.bark();    // "Max says Woof!"

        // Objects have independent state
        buddy.age = 4;  // Only buddy's age changes
        System.out.println(max.age);  // Still 5!
    }
}
```

Java

```
# Class definition (Python)
class Dog:
    # Constructor (special method)
    def __init__(self, name, age):
        self.name = name  # self refers to the object
```

```
        self.age = age

    def bark(self):
        print(f"{self.name} says Woof!")

# Instantiation
buddy = Dog("Buddy", 3)
max = Dog("Max", 5)

buddy.bark()  # "Buddy says Woof!"
max.bark()    # "Max says Woof!"
```

Python

```
class Dog {
public:
    string name;   // Instance variables
    int age;

    // Constructor
    Dog(string name, int age) : name(name), age(age) {}

    void bark() {
        cout << name << " says Woof!" << endl;
    }
};

int main() {
    Dog* buddy = new Dog("Buddy", 3);  // Heap allocation
    Dog max("Max", 5);                  // Stack allocation

    buddy->bark();  // "Buddy says Woof!"
    max.bark();     // "Max says Woof!"

    delete buddy;   // Don't forget to free heap memory!
}
```

C++

The `new` Keyword (Instantiation)

```
Dog buddy = new Dog(...);
```

What `new` does: 1. Allocates memory for the object (usually on the heap) 2. Initializes fields to default values (0, null, false) 3. Calls the constructor method 4. Returns a reference to the object

`new Dog("Buddy", 3)`

```
      |
      v
+--------------------------------+
| 1. Allocate memory (heap)      |
| 2. Initialize fields:          |
|    name = null                 |
|    age = 0                     |
| 3. Call constructor:           |
|    name = "Buddy"              |
|    age = 3                     |
| 4. Return reference (address)  |
+--------------------------------+
```

Default Values

```java
class Example {
    int num;           // Default: 0
    double decimal;    // Default: 0.0
    boolean flag;      // Default: false
    String text;       // Default: null
    int[] arr;         // Default: null
    Object obj;        // Default: null

    // Constructor to initialize
    Example() {
        num = 10;  // Now it's 10
        // text still null if not initialized
    }
}
```

Default values table:

Type	Default Value
byte, short, int, long	0
float, double	0.0
char	‘ŏ000’ (null character)
boolean	false
Object reference	null

Memory Where Objects Live

```
STACK (local variables)          HEAP (objects)
+---------------------+         +---------------------+
| Dog buddy ----------+-------->| Dog object          |
| (reference)         |         | name: "Buddy"       |
+---------------------+         | age: 3              |
                                +---------------------+
```

In Java, Python, C#: - **Objects live on the Heap** (managed memory) - **References live on the Stack** (variables point to heap)

In C++: - Objects can live on **Stack** (if declared locally) or **Heap** (if `new`)

Edge Cases & Traps

Trap 1: Reference vs Object Confusion

```
Dog d1 = new Dog("Buddy", 3);
Dog d2 = d1;  // Copies REFERENCE, not object!

// Now d1 and d2 point to SAME object!
d2.name = "Max";
System.out.println(d1.name);  // "Max"! Changed because same object!

// For independent copies, need to copy explicitly:
Dog d3 = new Dog(d1.name, d1.age);  // Creates NEW object
```

Trap 2: Null Reference

```
Dog d;  // d is null (no object!)
d.bark();  // NullPointerException! Object doesn't exist!

// Must initialize first:
d = new Dog("Buddy", 3);
d.bark();  // Now works
```

Trap 3: Uninitialized Fields

```
class Example {
    int x;  // Default 0, not uninitialized garbage!
}
// x is 0, not random garbage
// Unlike C/C++ local variables!
```

Trap 4: Class is a Type, Object is an Instance

```
Dog d = new Dog();  // Dog is type/class, d is instance
// You cannot do: new Dog.bark()  (wrong!)
// Must: d.bark() (using object instance)
```

Examples

```
class BankAccount {
    // Fields
    String accountNumber;
    String ownerName;
    double balance;

    // Constructor
```

```
    BankAccount(String accNum, String name, double initialBalance) {
        accountNumber = accNum;
        ownerName = name;
        balance = initialBalance;
    }

    // Methods
    void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
        }
    }

    void withdraw(double amount) {
        if (amount <= balance) {
            balance -= amount;
        } else {
            System.out.println("Insufficient funds!");
        }
    }

    void display() {
        System.out.println(ownerName + ": $" + balance);
    }
}

public class Main {
    public static void main(String[] args) {
        BankAccount acc1 = new BankAccount("12345", "Alice", 1000);
        BankAccount acc2 = new BankAccount("67890", "Bob", 500);

        acc1.deposit(500);   // Alice: $1500
        acc2.withdraw(100);  // Bob: $400
        acc1.display();       // Alice: $1500.0
        acc2.display();       // Bob: $400.0

        // Each object has its OWN state
        System.out.println(acc1.balance);  // 1500.0
        System.out.println(acc2.balance);  // 400.0
    }
}
```

Example 1: Bank Account Class

```
class Rectangle:
    def __init__(self, width, height):
```

```
        self.width = width
        self.height = height

    def area(self):
        return self.width * self.height

    def perimeter(self):
        return 2 * (self.width + self.height)

    def scale(self, factor):
        self.width *= factor
        self.height *= factor

# Create two rectangle objects
r1 = Rectangle(5, 3)
r2 = Rectangle(4, 2)

print(f"r1 area: {r1.area()}")  # 15
print(f"r2 area: {r2.area()}")  # 8

r1.scale(2)  # Scale r1 by 2
print(f"r1 area after scale: {r1.area()}")  # 60
print(f"r2 unchanged: {r2.area()}")  # Still 8!
```

Example 2: Rectangle Class

```
class Student {
    String name;
    int age;
    String major;

    // Constructor 1: Full
    Student(String name, int age, String major) {
        this.name = name;
        this.age = age;
        this.major = major;
    }

    // Constructor 2: Default major
    Student(String name, int age) {
        this.name = name;
        this.age = age;
        this.major = "Undeclared";
    }

    // Constructor 3: Default values
```

```
    Student() {
        this.name = "Unknown";
        this.age = 0;
        this.major = "Undeclared";
    }
}

// Usage
Student s1 = new Student("Alice", 20, "CS");          // Full
Student s2 = new Student("Bob", 19);                   // Default major
Student s3 = new Student();                            // All defaults
```

Example 3: Constructor Overloading (Multiple Constructors)

The `this` Reference

The `this` keyword is a reference to the **current object** - the object on which the method is being called.

```
class Person {
    private String name;
    private int age;

    // this helps distinguish field from parameter
    Person(String name, int age) {
        this.name = name;   // this.name = field, name = parameter
        this.age = age;     // this.age = field, age = parameter
    }

    // this can be used to return the object itself (for chaining)
    Person setName(String name) {
        this.name = name;
        return this;  // Returns the same object
    }

    Person setAge(int age) {
        this.age = age;
        return this;
    }

    // Chain method calls
    public static void main(String[] args) {
        Person p = new Person("Alice", 20);
        p.setName("Bob").setAge(25);  // Method chaining!
    }
}
```

Common uses of `this`:

Use	Example	When Needed
Disambiguate field/parameter	`this.x = x`	Parameter has same name as field
Return current object	`return this`	For method chaining
Pass current object	`callback(this)`	When object needs to reference itself
Constructor chaining	`this(args)`	Call another constructor in same class

this() for Constructor Chaining:

```
class Student {
    String name;
    int age;
    String major;

    // Chain to the full constructor
    Student(String name, int age) {
        this(name, age, "Undeclared");  // Calls Student(String, int, String)
    }

    Student(String name, int age, String major) {
        this.name = name;
        this.age = age;
        this.major = major;
    }
}
```

Important Note: `this()` must be the **first statement** in constructor!

Immutability

An **immutable object** is an object whose state **cannot be changed** after creation.

```
// Immutable class example
final class Person {
    private final String name;  // final = can't reassign
    private final int age;

    // No setters! Only getters and constructor
    Person(String name, int age) {
        this.name = name;   // Can't change after construction
        this.age = age;
    }

    // Only getters, no setters
    public String getName() { return name; }
    public int getAge() { return age; }
}

// Usage
```

```java
Person p = new Person("Alice", 25);
String n = p.getName();  // Can read, can't modify
// p.setAge(26);         // ERROR! No setter method exists!
```

Why Immutability Matters:

```
Benefits of Immutable Objects:
1. Thread-safe - No synchronization needed!
2. Cacheable - Can safely reuse instances
3. No defensive copies needed
4. Building blocks for reliable code

Examples of immutable types:
- String in Java
- int, double primitives
- LocalDate in Java 8
- tuple in Python
```

Creating Immutable Classes:

```java
public final class ImmutablePoint {
    private final int x;
    private final int y;

    // No setters! Only constructor and getters
    public ImmutablePoint(int x, int y) {
        this.x = x;
        this.y = y;
    }

    public int getX() { return x; }
    public int getY() { return y; }

    // Returns NEW point, doesn't modify this
    public ImmutablePoint translate(int dx, int dy) {
        return new ImmutablePoint(x + dx, y + dy);
    }
}

// Usage
ImmutablePoint p1 = new ImmutablePoint(1, 2);
ImmutablePoint p2 = p1.translate(3, 4);  // Creates NEW point, p1 unchanged
```

The Problem with Mutable Objects:

```java
// Mutable version - problematic!
class MutablePoint {
    int x, y;
    MutablePoint(int x, int y) { this.x = x; this.y = y; }
```

```
    void setX(int x) { this.x = x; }
    void setY(int y) { this.y = y; }
}

// Problem: If stored in collection, can be modified from outside!
MutablePoint p = new MutablePoint(1, 2);
List<MutablePoint> points = new ArrayList<>();
points.add(p);
points.get(0).setX(999);  // p.x is now 999! Original modified!

// Immutable version - safe!
ImmutablePoint p1 = new ImmutablePoint(1, 2);
points.add(p1);
// No way to modify p1 after creation!
```

1.2 Comparing and Copying Objects

Comparing Objects

Reference Equality vs Value Equality

Type	What It Checks	Syntax
Reference equality	Same memory location?	`==` in Java for references
Value equality	Same content?	`.equals()` in Java

```
String s1 = new String("hello");
String s2 = new String("hello");

// Reference comparison (are they the same object?)
System.out.println(s1 == s2);           // false (different objects)

// Value comparison (do they have same content?)
System.out.println(s1.equals(s2));      // true (same content!)

// BUT: String literals are pooled!
String s3 = "hello";
String s4 = "hello";

System.out.println(s3 == s4);           // true (same pooled object!)
System.out.println(s3.equals(s4));     // true (same content too)
```

```
class Point {
    int x, y;
    Point(int x, int y) { this.x = x; this.y = y; }
```

```
}

Point p1 = new Point(1, 2);
Point p2 = new Point(1, 2);

System.out.println(p1 == p2);       // false! Different objects!
System.out.println(p1.equals(p2)); // false! equals() inherited from Object
                                    // does reference comparison by default!

// You need to OVERRIDE equals()!
class Point {
    int x, y;
    Point(int x, int y) { this.x = x; this.y = y; }

    @Override
    public boolean equals(Object obj) {
        if (this == obj) return true;  // Same reference
        if (!(obj instanceof Point)) return false;
        Point other = (Point) obj;
        return this.x == other.x && this.y == other.y;
    }
}
```

Why == Doesn't Work for Value Comparison

```
# Python: == does VALUE comparison by default!
class Point:
    def __init__(self, x, y):
        self.x = x
        self.y = y

p1 = Point(1, 2)
p2 = Point(1, 2)
p3 = p1

print(p1 == p2)  # True! (Python's == compares values by default)
print(p1 == p3)  # True! (Same object, same value)

# is checks identity (like reference equality in Java)
print(p1 is p2)  # False! Different objects
print(p1 is p3)  # True! Same object
```

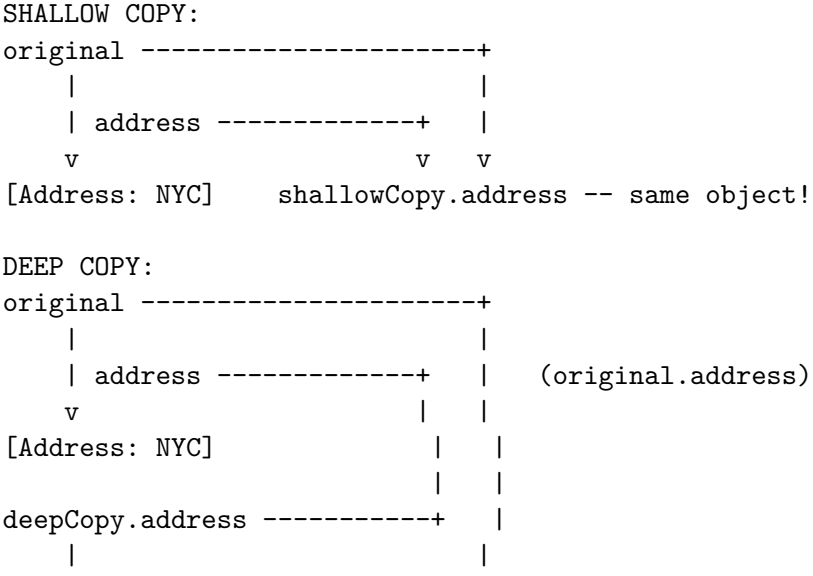
Python Comparison

Copying Objects

```java
class Address {
    String city;
    Address(String city) { this.city = city; }
}

class Person {
    String name;
    Address address;  // Reference type field!

    Person(String name, Address address) {
        this.name = name;
        this.address = address;
    }
}

// Original
Address addr = new Address("NYC");
Person original = new Person("Alice", addr);

// Shallow copy - copies reference, not object!
Person shallowCopy = original;
// shallowCopy.address points to SAME Address object as original!

// Deep copy - copies the referenced objects too!
Person deepCopy = new Person(original.name, new Address(original.address.city));
// deepCopy.address is a NEW object with same values!
```

Shallow Copy vs Deep Copy

Shallow vs Deep Copy Visualization



```
    v                           v
[Address: NYC]              NEW COPY! Different object
```

```
class Person {
    String name;
    int age;

    // Regular constructor
    Person(String name, int age) {
        this.name = name;
        this.age = age;
    }

    // Copy constructor
    Person(Person other) {
        this.name = other.name;
        this.age = other.age;
    }
}

// Using copy constructor
Person original = new Person("Alice", 25);
Person copy = new Person(original);  // Creates NEW object with same values!
```

Copy Constructors

```
import copy

class Person:
    def __init__(self, name, address):
        self.name = name
        self.address = address  # Could be object reference

# Original
p1 = Person("Alice", "NYC")

# Shallow copy - nested objects shared
p2 = copy.copy(p1)

# Deep copy - nested objects also copied
p3 = copy.deepcopy(p1)

# p1 and p2 are different objects, but their address field might be same object
# p3 is completely independent
```

Python Copy Module

Edge Cases & Traps

Trap 1: == on Objects Compares References, Not Values

```
String a = new String("test");
String b = new String("test");
if (a == b) {  // FALSE! Different objects!
    // This block never executes!
}
if (a.equals(b)) {  // TRUE! Same content!
    // This block executes!
}
```

[WARNING] **Professor Trap**: Always use `.equals()` for string value comparison!

Trap 2: Mutable Default Arguments (Python)

```
# DANGER! This is a common Python trap!
class BadList:
    def __init__(self, items=[]):  # Mutable default!
        self.items = items

lst1 = BadList()
lst1.items.append(1)

lst2 = BadList()
print(lst2.items)  # [1]! Shared default list!

# CORRECT:
def __init__(self, items=None):
    if items is None:
        items = []  # Create new list each time
```

Trap 3: Shallow Copy with Nested Objects

```
class Node {
    Node next;
    int value;
}

Node n1 = new Node();
n1.value = 1;
Node n2 = new Node();
n2.value = 2;
n1.next = n2;

// Shallow copy of n1
Node copy = new Node();
copy.value = n1.value;
copy.next = n1.next;  // copy.next points to SAME n2!
```

```
// Now modifying n2 affects both original and copy!
```

Trap 4: Forgetting to Override equals()

```
class Point {
    int x, y;
    // No equals() override!
}

Point p1 = new Point(1, 2);
Point p2 = new Point(1, 2);

// These are both false!
System.out.println(p1 == p2);      // false (different objects)
System.out.println(p1.equals(p2)); // false (inherited == checks references!)
```

[WARNING] **Professor Trap**: If you override equals(), also override hashCode() for Maps/Sets!

Examples

```
class Point {
    private int x, y;

    public Point(int x, int y) {
        this.x = x;
        this.y = y;
    }

    @Override
    public boolean equals(Object obj) {
        if (this == obj) return true;  // Same reference
        if (obj == null || getClass() != obj.getClass()) return false;
        Point other = (Point) obj;
        return x == other.x && y == other.y;
    }

    @Override
    public int hashCode() {
        return 31 * x + y;  // Consistent with equals()
    }
}

// Now:
Point p1 = new Point(1, 2);
Point p2 = new Point(1, 2);
System.out.println(p1.equals(p2));  // true! Same values!
System.out.println(p1 == p2);       // false! Different objects!
```

Example 1: Proper equals() Implementation

```
class Person implements Cloneable {
    String name;
    int age;

    Person(String name, int age) {
        this.name = name;
        this.age = age;
    }

    @Override
    protected Object clone() {
        return new Person(this.name, this.age);  // Deep copy
    }
}

Person original = new Person("Alice", 25);
Person cloned = (Person) original.clone();

System.out.println(original == cloned);       // false
System.out.println(original.name.equals(cloned.name));  // true
```

Example 2: Clone Method

1.3 Testing

Testing Concepts

Why test? - Find bugs before users do - Verify functionality works correctly - Enable confident refactoring - Document expected behavior

Unit Testing Basics

```
// Simple unit test example (JUnit)
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

class Calculator {
    int add(int a, int b) { return a + b; }
    int divide(int a, int b) {
        if (b == 0) throw new ArithmeticException("Division by zero");
        return a / b;
    }
}
```

```java
class CalculatorTest {
    @Test
    void testAdd() {
        Calculator calc = new Calculator();
        assertEquals(5, calc.add(2, 3));
        assertEquals(0, calc.add(-1, 1));
    }

    @Test
    void testDivide() {
        Calculator calc = new Calculator();
        assertEquals(2, calc.divide(10, 5));
    }

    @Test
    void testDivideByZero() {
        Calculator calc = new Calculator();
        assertThrows(ArithmeticException.class, () -> calc.divide(1, 0));
    }
}
```

Testing Terminology

Term	Meaning
Unit test	Tests single method/class
Integration test	Tests multiple components together
Test fixture	Setup code before tests
Assertion	Check that expected == actual
Mock	Fake object for testing dependencies
TDD	Test-Driven Development (write tests first)

White-Board Example: Testing OOP Concepts

```
Question: What is the output?

Dog d1 = new Dog("Buddy", 3);
Dog d2 = d1;           // What is d2?
d2.setAge(4);
System.out.println(d1.getAge());

Answer: 4

Explanation:
1. d1 created with age 3
2. d2 = d1 means d2 points to SAME object as d1
3. d2.setAge(4) modifies the shared object
```

4. d1.getAge() returns 4

This tests understanding of REFERENCE semantics!

Practice Questions

MCQ 1: What does `new` do in Java? - A) Declares a new variable - B) Allocates memory and calls constructor - C) Copies an existing object - D) Deletes an object

MCQ 2: Two objects created with `new Dog()` and same values - what is `==`? - A) true - B) false - C) Depends on equals() method - D) Compile error

MCQ 3: What is shallow copy? - A) Copies all fields including references - B) Copies only primitive fields - C) Creates new referenced objects - D) Used for value types only

Short Answer: Explain why `String s1 = new String("hi") == String s2 = new String("hi")` returns false.

Board Coding: Write a class `Circle` with radius field, constructor, and method `getArea()`. Then write tests for it.

Oral Explanation: What's the difference between `==` and `.equals()` for objects? When would you use each?

Solutions

MCQ 1: B) Allocates memory and calls constructor

MCQ 2: B) false - different objects in memory

MCQ 3: A) Copies all fields, but reference fields point to same objects

Short Answer: `new` creates a new object in heap memory. Even though both strings have content "hi", they are different objects at different memory locations, so `==` compares references (addresses) and returns false. Use `.equals()` for content comparison.

Cheat-Sheet

Concept	Key Points
Class	Blueprint/template defining structure and behavior
Object	Instance of class, lives in heap memory
Instantiation	`new ClassName()` allocates and initializes
Constructor	Initializes new object, has same name as class
Reference	Variable that points to object in heap
null	Reference pointing to nothing
==	Compares references (memory addresses)
equals()	Compares object values (override for custom comparison)
Shallow copy	Copies object, shared references inside
Deep copy	Copies object AND all nested objects

Review Checklist

- ☐ **Class vs Object**: Can you explain the difference?
- ☐ **Instantiation**: Do you understand what `new` does?
- ☐ **Reference semantics**: Can you trace reference assignments?
- ☐ **Comparison**: Do you know when to use == vs equals()?
- ☐ **Copying**: Can you explain shallow vs deep copy?
- ☐ **Testing**: Can you write basic unit tests?